

Introduction to Relation

Lecture-7

Closures of Relations



Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Def: Let R be a relation on a set A that may or may not have some property P . (Ex: Reflexive,...) If there is a relation S with property P containing R such that S is a subset of every relation with property P containing R , then S is called the closure of R with respect to P .

Note: the closure may or may not exist

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Theorem: Let R be a relation on A . The **reflexive closure** of R , denoted $r(R)$, is $R \cup \Delta$.

Add loops to all vertices on the digraph representation of R .

- Put 1's on the diagonal of the connection matrix of R .

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Example 1: What is the reflexive closure of the relation $R = \{(a, b) \mid a < b\}$ on the set of integers?

Example 2: What is the symmetric closure of the relation $R = \{(a, b) \mid a > b\}$ on the set of positive integers?

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Definition: Let R be a relation on A . Then R^{-1} or the *inverse* of R is the relation $R^{-1} = \{ \langle y, x \rangle \mid \langle x, y \rangle \in R \}$

Note: to get R^{-1}

- reverse all the arcs in the digraph representation of R
- take the transpose M^T of the connection matrix M of R .

Note: This relation is sometimes denoted as R^T or R^c and called the *converse* of R

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

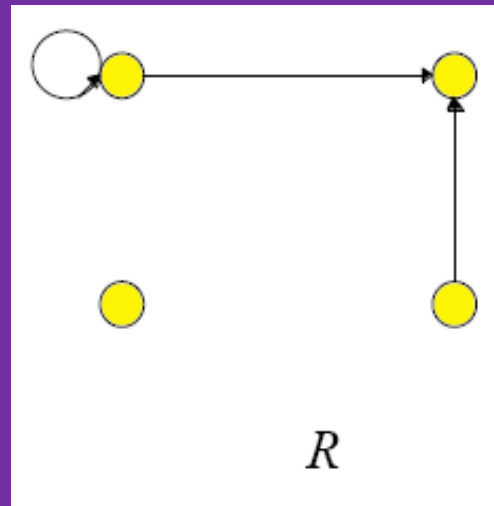
Theorem: Let R be a relation on A . The **symmetric closure** of R , denoted $s(R)$, is the relation $R \cup R^{-1}$.

Example 2: What is the symmetric closure of the relation $R = \{(a, b) \mid a > b\}$ on the set of positive integers?

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

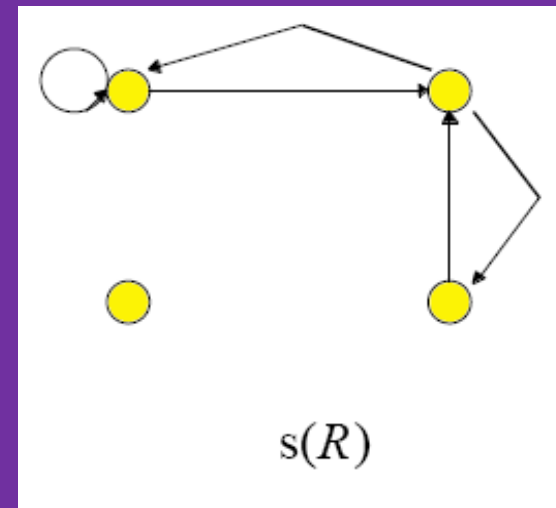
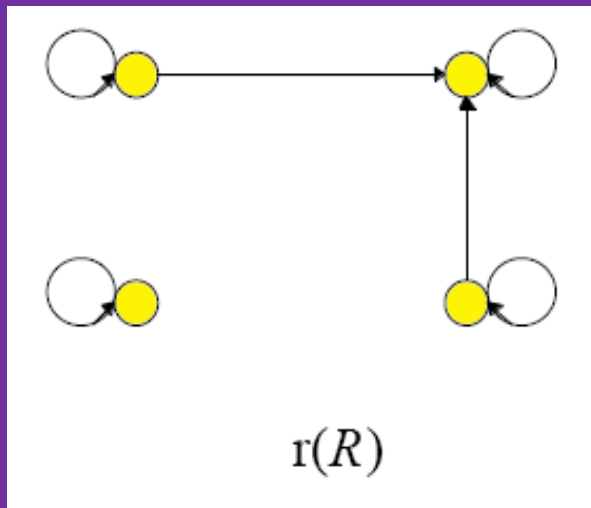
What is the Reflexive and symmetric closure of



Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

The Reflexive and symmetric closure of



Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Transitive Closure

- Recall that the transitive closure of a relation R , $t(R)$, is the smallest transitive relation containing R .
- Also recall: R is transitive **iff** R^n is contained in R for all n .
- Hence, if there is a path from x to y then there must be an arc from x to y , or $\langle x, y \rangle$ is in R .

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

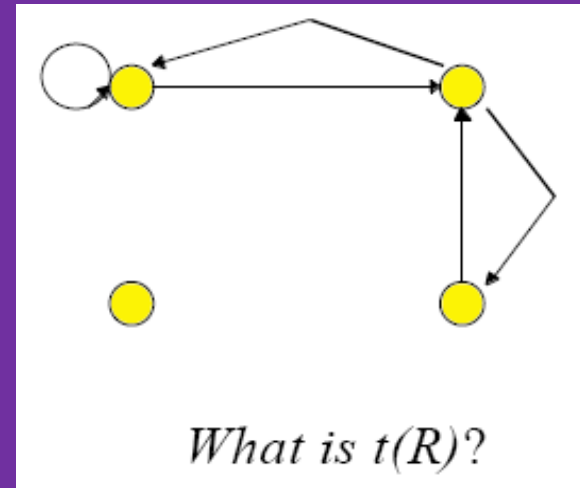
Transitive Closure

- Recall that the transitive closure of a relation R , $t(R)$, is the smallest transitive relation containing R .
- Also recall: R is transitive **iff** R^n is contained in R for all n .
- Hence, if there is a path from x to y then there must be an arc from x to y , or $\langle x, y \rangle$ is in R .

Closers of relation
Warshall's Algorithm to calculate Transitive Closure

Introduction

What is Transitive Closure



Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Theorem : Let M_R be the zero-one matrix of the relation R on the relation R on a set with n elements. Then the zero-one matrix of the transitive closure R^* is

$$M_{R^*} = M_R \vee M_R^{[2]} \vee M_R^{[3]} \vee \cdots \vee M_R^{[n]}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Example: Find the zero-one matrix of the transitive closure of the relation R where

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive , symmetric, transitive closure of R using.

- (i) Composition of relation R
- (ii) Composition of matrix relation R
- (iii) Graphical representation of R

Closers of relation
Warshall's Algorithm to calculate Transitive Closure

Introduction

Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive , symmetric, transitive closure of R using.

(i) Composition of relation R

- Reflexive Closure: $S=R \cup \Delta = \{(1,2),(2,3),(3,1)\} \cup \{(1,1), (2,2), (3,3)\}$
- $S=\{(1,1), (1,2), (2,2), (2,3), (3,1), (3,3)\}$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive , symmetric, transitive closure of R using.

(i) Composition of relation R

- Symmetric Closure: $S=R \cup R^{-1} = \{(1,2),(2,3),(3,1)\} \cup \{(2,1), (3,2), (1,3)\}$
- $S=\{(2,1), (1,2), (3,2), (2,3), (3,1), (1,3)\}$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive , symmetric, transitive closure of R using.

(i) Composition of relation R

- Transitive Closure: $S=R \cup R^2 \cup R^3 =$
- $\{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

Closers of relation
Warshall's Algorithm to calculate Transitive Closure

Introduction

Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive, symmetric, transitive closure of R using.

(ii) Composition of relation on Matrix

- Reflexive Closure: $S = M \cup I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \cup \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- $= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \therefore S =$

$\{(1,1), (1,2), (2,2), (2,3), (3,1), (3,3)\}$

Closers of relation Warshall's Algorithm to calculate Transitive Closure

Introduction

Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive , symmetric, transitive closure of R using.

(i) Composition of relation on Matrix

Transitive Closure : $S=M \cup M^2 \cup M^3$

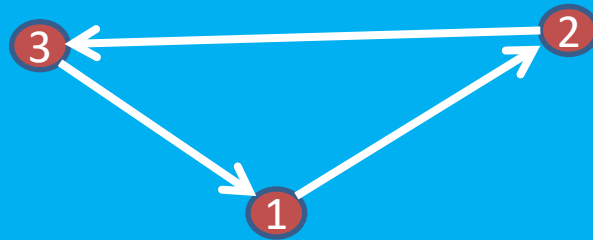
$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \cup \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \cup \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\bullet = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \therefore S = \{(1,1)(1,2), (1,3), (2,1), (2,2)(2,3), (3,1), (3,2), (3,3)\}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

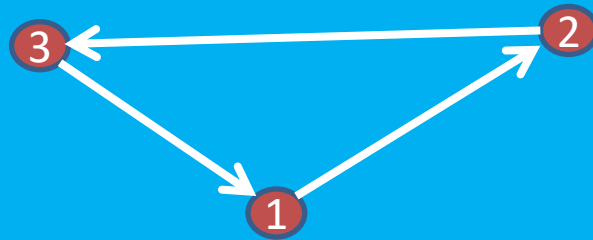
Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive , symmetric, transitive closure of R using.
(iii) Composition of relation on Di-Graph



Closers of relation
Warshall's Algorithm to calculate Transitive Closure

Introduction

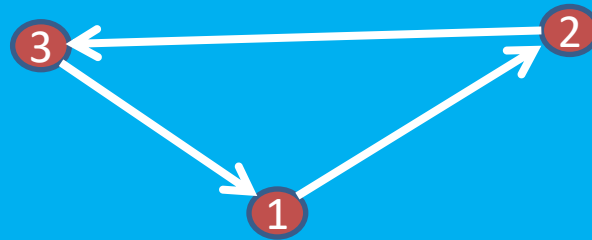
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Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

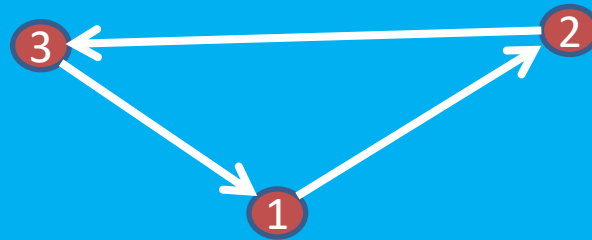
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Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Example: Let $R=\{(1,2),(2,3),(3,1)\}$ and $A=\{1,2,3\}$ find reflexive , symmetric, transitive closure of R using.
(iii) Composition of relation on Di-Graph



Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

Warshall's Algorithm to calculate Transitive Closure

$R^K[i, j] = 1$
if $R^{K-1}[i, j]$ is 1 or $R^{K-1}[i, K]$ and $R^{K-1}[K, j]$ is 1

Closers of relation
Warshall's Algorithm to calculate Transitive Closure

Introduction

By using Warshall's Algorithm calculate Transitive Closure of relation R on $A=\{1,2,3\}$ where $R=\{(1,1), (1,3), (2,2), (3,1), (3,2)\}$

$$R^0 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

$$R^0 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R^1[i, j] = 1$$

if $R^0[i, j]$ is 1 or $R^0[i, 1]$ and $R^0[1, j]$ is 1

$$R^1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

$$R^1 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R^2[i, j] = 1$$

if $R^1[i, j]$ is 1 or $R^1[i, 2]$ and $R^1[2, j]$ is 1

$$R^2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

$$R^2 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$R^3[i, j] = 1$$

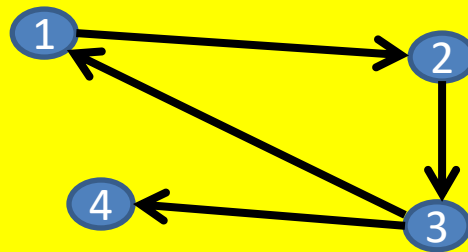
if $R^2[i, j]$ is 1 or $R^2[i, 3]$ and $R^1[3, j]$ is 1

$$R^3 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive Closure

Introduction

By using Warshall's Algorithm calculate Transitive Closure of relation R on $A=\{1,2,3,4\}$ where



$$R^0 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

$$R^0 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^1[i, j] = 1$$

if $R^0[i, j]$ is 1 or $R^0[i, 1]$ and $R^0[1, j]$ is 1

$$R^1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

$$R^1 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^2[i, j] = 1$$

if $R^1[i, j]$ is 1 or $R^1[i, 2]$ and $R^1[2, j]$ is 1

$$R^2 = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

$$R^2 = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^3[i, j] = 1$$

if $R^2[i, j]$ is 1 or $R^2[i, 3]$ and $R^2[3, j]$ is 1

$$R^3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Closers of relation
Warshall's Algorithm to calculate Transitive
Closure

Introduction

$$R^3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$R^4[i, j] = 1$
if $R^3[i, j]$ is 1 or $R^3[i, 4]$ and $R^3[3, 4]$ is 1

$$R^4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$